Lab 4: Understanding Multipath Propagation in Wireless Communication

Part 1: 多径信道的建模和观察
Modelling and Observing a Multipath Channel

Step 1 – Transmitted Signal Preparation 传输信号准备

Generate a transmitted signal of your choice (e.g., sinusoidal carrier, BPSK/QPSK baseband, or OFDM symbol) that will pass through a wireless channel. Define the number of propagation paths realistically (e.g., 3–6 paths) and specify:

- The attenuation of each path
- The time delay of each path
- Whether delays are constant or time-varying

You should justify your chosen values based on a real-world scenario (e.g., indoor room, urban street, moving vehicle, etc.).

Step 2 – Received Signal Under Multipath 多径下接收信号

Model the received signal as the superposition of multiple delayed and attenuated versions of the transmitted signal.

$$y(t) = \sum_{j=1}^{J} \beta_j(t)\, x(t - \tau_j(t))$$

Where:

- J= number of multipath components
- $\beta_j(t)$= attenuation of j^{th}path (may vary with time)
- $\tau_j(t)$= time delay of j^{th}path (may vary with time)

Observe and compare the transmitted and received signal in the time domain. Record any distortion, spreading, or amplitude fluctuation caused by multipath.

Step 3 – Transfer Function of Multipath Channel 多径信道的传递函数

For a given input frequency f_0, derive and plot the **time-varying channel transfer function**:

$$H(f_0, t) = \sum_{j=1}^{J} \beta_j(t)\, e^{-j2\pi f_0 \tau_j(t)}$$

Repeat this for **several time instances** to show how the channel frequency response changes over time.

Plot the magnitude response $|H(f,t)|$ versus frequency.

Step 4 – Impulse Response of the Multipath Channel 多径信道的脉冲响应

Using the derived channel parameters, compute and plot the **impulse response**:

$$h(t,\tau) = \sum_{j=1}^{J} \beta_j(t)\,\delta(\tau - \tau_j(t))$$

Plot the impulse response for **at least 4 different time instants** to demonstrate how the channel taps evolve in time (slow fading or fast fading depending on your chosen model).

Tasks (Part 1)

1. Plot both the transfer function and the impulse response of the multipath channel for multiple time samples.
2. Compare the time-domain received signal with the original transmitted signal and comment on ISI (Inter-Symbol Interference).
3. Provide a short conclusion summarizing what the plots show about multipath effects (delay spread, frequency selectivity, time variation).

1. 绘制多个时间点下信道频率响应和冲激响应。

2. 对比时域接收信号与发送信号，分析符号间干扰（ISI）。

3. 简要总结图表中涉及的多径影响，如延迟扩展、频率选择性、时变特性。

了解多径无线信道中的衰落

Part 2: Understanding Fading in Multipath Wireless Channels

Wireless channels may exhibit Rayleigh fading (no direct path) or Rician fading (dominant LOS path). These are modelled statistically. Figure 4.1 illustrates these two different environments: one for LOS and the other for NLOS.

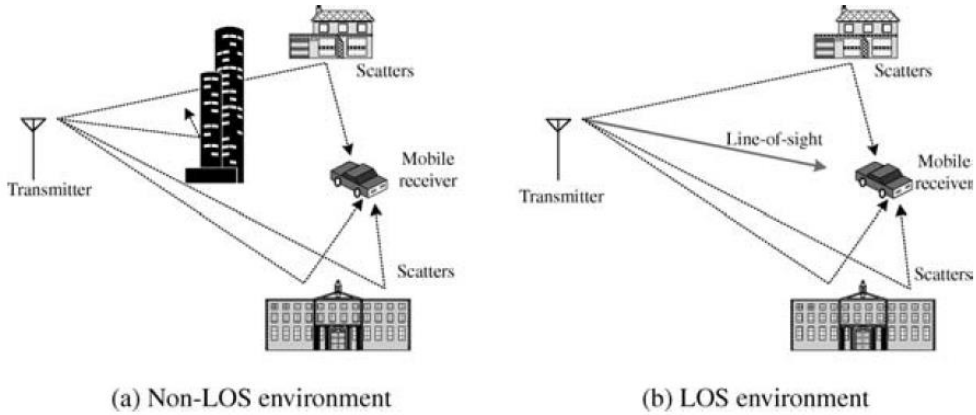

Figure 4.1: Non-LOS and LOS propagation environments.

Step 1 – Rayleigh Fading Model 瑞利衰落模型

Assume no line-of-sight (NLOS).

The amplitude of the received signal follows the Rayleigh distribution:

$$p(x) = \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}}, x \geq 0$$

Simulate and plot the amplitude distribution of the Rayleigh channel. 模拟并绘制瑞利通道的幅度分布

Step 2 – Rician Fading Model 莱斯衰落模型

Assume LOS + scattered multipath.

The amplitude follows the Rician distribution:

$$p(x) = \frac{x}{\sigma^2} e^{-\frac{x^2+A^2}{2\sigma^2}} I_0(\frac{Ax}{\sigma^2})$$

where Arepresents the dominant LOS component, and $K = \frac{A^2}{2\sigma^2}$is the Rician K-factor.

Simulate Rician fading for at least two different K-factors (e.g., K = 3 dB and K = 10 dB).
模拟至少两个不同 K 因子的莱斯衰落

Tasks (Part 2)

1. Plot Rayleigh and Rician fading amplitude distributions on the same graph as shown in Fig 4.2 (Make any realistic assumptions if required).

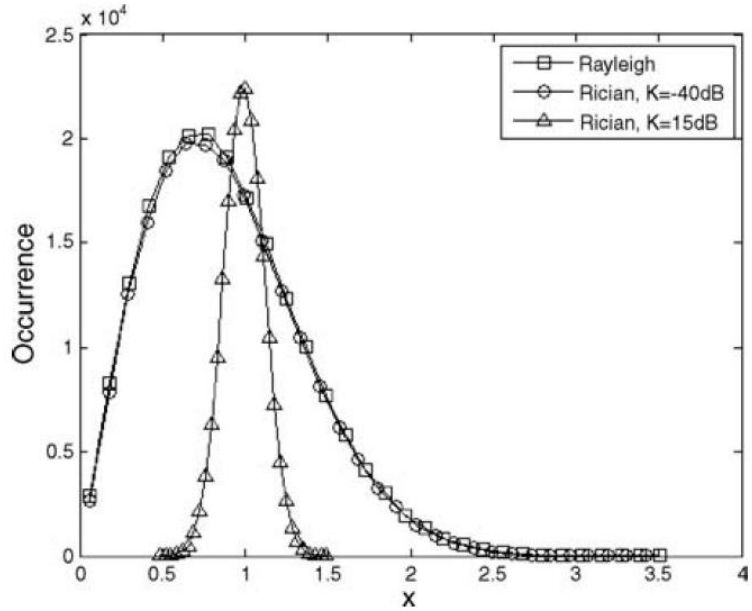

Figure 4.2: Distributions for Rayleigh and Rician fading channels.

2. Explain the physical meaning of the K-factor and how Rician reduces to Rayleigh when K = 0 dB.

3. Write a short comparison: Which type of fading is typical for indoor, urban, or satellite links?

1. 在同一张图上绘制Rayleigh与Rician幅度分布曲线。

2. 解释K因子的物理意义，以及K=0 dB时Rician分布如何退化为Rayleigh分布。

3. 简要比较不同场景下主要发生哪种衰落：室内、城市、卫星链路等。

Discussion (Required in Report)

At the end of your lab, you must write a short discussion answering the following questions based on the results you observed in your figures, plots, and simulations. **When answering, do not just repeat theory from the lecture — you must connect each answer to something you saw in your own results (plots, measurements, effects, BER curves, etc.).**

Questions to Answer in Your Discussion

1. What aspects of multipath did you observe?
   您观察到了多径特征的哪些方面？
   - Refer to what you saw in the received signal, impulse response, or frequency response plots.
   - Mention effects such as signal distortion, amplitude changes, spreading, echoes, etc.
2. How does delay spread relate to ISI? 延迟扩展与 ISI 有何关系？
   - Use your eye diagram, waveform comparison, or BER result to show how ISI changed when delay spread increased.
   - Explain using *your* simulation, not just theory.
3. How does fading differ from deterministic multipath? 衰落与确定性多径有何不同？
   - Compare your Rayleigh/Rician results with the fixed multi-tap channel.
   - Mention if the fading signal was time-varying, random, more unpredictable, etc.
4. What would happen if symbol duration is shorter than delay spread?
   如果符号持续时间短于延迟扩展会发生什么？
   - Support your answer using ISI observations or BER performance.
   - You may refer to what happened when you used a fast symbol rate.
5. What wireless systems are most affected by Rayleigh fading? 哪些无线系统受瑞利衰落影响最大？
   - Answer using logic + what you observed in fading simulation.
   - Example: "Because Rayleigh fading produced deep nulls, systems without LOS like indoor Wi-Fi or street-level cellular links are most affected."